SCHOOL OF MATERIALS SCIENCE & ENGINEERING

CONTINUOUS ASSESSMENT 1 SEMESTER 1 2023-2024

MS1016 – THERMODYNAMICS OF MATERIALS

(Question Sheet)

26 September 2023 Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains **THREE (3)** short questions and **TEN (10)** MCQ questions.
2. Answer **ALL** questions.
3. Write all answers on the answer sheet provided and clearly write your name and matriculation number on it.
4. This is an open-notes test.
5. All questions have ONLY ONE correct answer for the MCQ questions.
6. You may retain this question sheet. Only answers on the submitted answer sheet will be considered.

PART I – SHORT QUESTIONS

(1) Show that $dH = TdS + VdP$ for a system that does work reversibly.
(20 marks)

Solution:

Given $$H = U + PV$$

$$dH = dU + PdV + VdP$$

From First Law of Thermodynamics,

$$dU = \delta Q - PdV$$

And for a reversible process, the Second Law of Thermodynamics is

$$dS = \delta Q/T$$

$$\therefore\ dH = \delta Q - PdV + PdV + VdP$$

$$dH = TdS + vdP$$

2. Thor, the God of Thunder, has taken a brief sabbatical from Asgardian duties to study thermodynamics at NTU as your classmate in Prof. Leonardo's class. Intrigued by the lessons, Thor wonders if the principles of thermodynamics could be applied to Mjölnir, his legendary hammer. The core of Mjölnir is a closed thermodynamic system that can be described by the Helmholtz free energy F(T,V). Subsequently, he has discovered through Prof. Leonardo's lessons that his ability as a god to manipulate thunder actually changes the entropy within Mjölnir.

   a. Define the Helmholtz free energy as it pertains to Mjölnir. Derive its differential form and how it relates to the First law of thermodynamics. **(5 marks)**

   b. Starting from the Helmholtz free energy of Mjölnir, derive one of Maxwell's equations. Break down how this equation might allow Thor to understand the manipulation of entropy within Mjölnir when he uses his thunder powers. **(10 marks)**

   c. Relate the implications of a positive and a negative ΔF within Mjölnir on Thor's combat effectiveness. **(5 marks)**

Solutions:

(a) Definition of Helmholtz Free Energy and its Relevance to Thor (5 marks):
The Helmholtz Free Energy (F) is defined as:

$$F = U - TS$$

where (U) is the internal energy, (T) is the temperature, and (S\) is the entropy of Mjölnir. Taking its differential, we have:

$$dF = dU - TdS - SdT$$

Substituting ($dU = TdS - PdV$) from the First Law of Thermodynamics, we get:

$$dF = -SdT - PdV$$

Relevance to Thor: Thor can use this understanding to gauge how using thunder might change the internal energy and entropy within Mjölnir, affecting its battle capabilities.

(b) Derivation of Maxwell's Equation and Its Relevance to Thor (10 marks):

Starting from the differential form of (F):

$$dF = -SdT - PdV$$

Solution should identify the partial derivatives of F(T,V):

$$\left(\frac{\partial F}{\partial T}\right)_V = -S, \quad \left(\frac{\partial F}{\partial V}\right)_T = -P$$

Applying the mixed partial derivative theorem, we get:

$$\frac{\partial^2 F}{\partial T\,\partial V} = \frac{\partial^2 F}{\partial V\,\partial T}$$

Substituting the partial derivatives of (F):

$$-\left(\frac{\partial S}{\partial V}\right)_T = -\left(\frac{\partial P}{\partial T}\right)_V$$

Relevance to Thor: This equation shows that Thor's ability to manipulate thunder can influence the entropy and pressure within Mjölnir. For example, if Thor can control the temperature change via his thunder powers, he can also affect the pressure and entropy within the hammer, allowing for strategically stronger or more efficient attacks.

c) Implications of Changes in (ΔF) for Thor (5 marks):

A negative (ΔF) implies that Mjölnir has moved to a state where it has more "free energy" to do work, enabling Thor to unleash more potent attacks. Conversely, a positive (ΔF) suggests Mjölnir is losing its capability to perform useful work, potentially weakening Thor's strikes.

3. You are a resourceful scientist who has survived a shipwreck and along with a few others. Stranded on a deserted island with limited supplies, you realise that conventional cooking methods are not feasible. Amidst the wreckage, you find a chicken and ponder how to cook it. Having taken Prof. Leonardo's Introduction to Thermodynamics class, you remember a theoretical discussion about how Kinetic Energy can transfer heat and that you could potentially cook a chicken through slaps and decide to explore this unconventional method as a last resort. You make a series of assumptions: an average human hand mass of 0.4 kg, a slap velocity of 5 m/s at the point of impact, and an energy transfer efficiency of only 10%. The chicken weighs 1.5 kg, has a specific heat of 2.8 J/g°C, and needs to go from an initial temperature of 10°C to a safe-to-eat temperature of 75°C.

How many slaps do you need to cook the chicken?

Make sure you discuss possible limitations to your method. (20 marks)

Part 1: Calculate the kinetic energy imparted by a single slap based on the given assumptions. Provide the formula before giving your numerical answer. (4 marks)

$$KE = \frac{1}{2} \times \text{mass} \times \text{velocity}^2$$

$$KE = \frac{1}{2} \times 0.4\,\text{kg} \times (5\,\text{m/s})^2 = 5\,\text{J}$$

Part 2: Determine the effective energy transferred to the chicken from each slap considering the 10% efficiency. (4 marks)

$$\text{Effective Energy} = 0.10 \times 5\,\text{J} = 0.5\,\text{J} \text{ (4 marks)}$$

Part 3: Compute the total energy required to raise the chicken's temperature to 75°C. Provide the formula before giving your numerical answer. (4 marks)

$$\Delta Q = \text{mass} \times \text{Cp} \times \text{dT}$$

$$\Delta Q = 1500\,\text{g} \times 2.8\,\text{J/g°C} \times 65\,\text{°C} = 273{,}000\,\text{J}$$

Part 4: How many slaps will it take to cook the chicken to the desired temperature? Provide the formula before giving your numerical answer. (4 marks)

$$\text{Number of slaps} = \frac{\text{Total Energy required}}{\text{Effective Energy per slap}}$$

$$\text{Number of slaps} = \frac{273{,}000\,\text{J}}{0.5\,\text{J/slap}} = 546{,}000\,\text{slaps}$$

Part 5: Briefly discuss the limitations and assumptions of this unconventional cooking method. (4 marks)

1. The energy transfer efficiency is assumed to be 10%, which might not be accurate.
2. Uniform energy distribution throughout the chicken is assumed, which is unlikely.
3. The chicken is assumed to absorb all the effective energy, ignoring losses to the environment.
4. The slapper or the chicken may not survive so many slaps.

PART II MCQ QUESTIONS

Answer all the **TEN MCQ** questions below. There is only **ONE** answer for each question. (4 marks each)

1. "In a closed system where no energy is added or removed, the entropy of the system will not decrease." This statement is best described by:

A) First Law of Thermodynamics
B) Conservation of Mass
C) Third Law of Thermodynamics
D) Zeroth Law of Thermodynamics
E) Second Law of Thermodynamics

Explanation:

A) First Law of Thermodynamics - This law deals with the conservation of energy, not the directionality of processes.

B) Conservation of Mass - This principle is about the constancy of mass in a closed system and has nothing to do with entropy.

C) Third Law of Thermodynamics - This law states that the entropy of a perfect crystal is exactly zero when the temperature of the crystal is exactly zero. It doesn't deal with the change in entropy in a closed system.

D) Zeroth Law of Thermodynamics - This law provides the foundation for the concept of temperature, saying if two systems are each in thermal equilibrium with a third, they are in thermal equilibrium with each other.

E) Second Law of Thermodynamics - This law states that the entropy of an isolated system will tend to increase over time, approaching a maximum value at equilibrium. The statement given in the question is directly related to this principle, making E the correct answer.

2. The efficiency of a heat engine can be best improved by:

A) Lowering the temperature at which energy is received
B) Balancing the temperature between input and output energies, so $\Delta T = 0$
C) Raising the temperature at which energy is received
D) Lowering the operating temperature of the heat engine
E) Keeping the temperature of the heat engine constant

Explanation:

A) Lowering the temperature at which energy is received would generally reduce efficiency, as less thermal energy would be available to convert into work.

B) Balancing the temperature between input and output energies, so $\Delta T = 0$, would mean no work could be done, making the engine entirely inefficient.

C) Raising the temperature at which energy is received would generally improve efficiency, as higher thermal gradients allow for better energy conversion to work, making this the correct answer.

D) Lowering the operating temperature of the heat engine would not necessarily improve its efficiency and could potentially make it less efficient.

E) Keeping the temperature of the heat engine constant does not inherently improve efficiency.

3. A monoatomic ideal gas is contained in a cylinder that is in adiabatic conditions. The pressure of the gas drops from 30 atm to 10 atm, and the initial temperature is 400 K. What is the final temperature of the gas?

A) 205 K
B) 220 K
C) 226 K
D) 245 K
E) 260 K

Solution:

The formula $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{R/C_p}$ is given for this adiabatic process. For a monoatomic ideal gas, $C_p = \frac{5}{2}R$.

We have:

$$P_1 = 30\,\text{atm}$$

$$P_2 = 10\,\text{atm}$$
$$T_1 = 400\,\text{K}$$

Substituting these into the formula:

$$\frac{T_2}{400\,\text{K}} = \left(\frac{10\,\text{atm}}{30\,\text{atm}}\right)^{R/\frac{5}{2}R}$$

The (R) terms cancel out, leaving:

$$\frac{T_2}{400\,\text{K}} = \left(\frac{1}{3}\right)^{2/5}$$

$$\frac{T_2}{400\,\text{K}} = 0.33^{0.4}$$

$$\frac{T_2}{400\,\text{K}} \approx 0.6418\ (\text{approximately})$$

$$T_2 \approx 256.72\,\text{K}$$

Rounded to the nearest whole number, (T_2) is approximately 260 K.

4. Which of the following process may be described as the Joule-Thompson effect?

A)Isentropic

B)Isenthalpic
C)Isothermal
D)Isochoric
E) Isobaric

5. Consider the following processes for an ideal gas in a thermodynamic system:
   (a) Process where both temperature and volume remain constant.
   (b) Process that is reversible and adiabatic.
   (c) Process at constant pressure.
   (d) Process at constant volume.
   (e) Process at constant temperature.

   Match each process with its appropriate descriptor:

   Descriptors:
   A) Isothermal
   B) Isochoric
   C) Isentropic
   D) Isobaric
   E) Adiabatic

   Choose the lettered option that best matches the sequence (a,b,c,d,e):

   A) B, C, D, A, E
   B) B, C, D, E, A
   C) C, A, B, D, E
   D) D, A, B, C, E
   E) B, C, D, B, A

Solution:

(a) Process where both temperature and volume remain constant: In this case, volume is constant, making it an isochoric process.
Answer for (a): B) Isochoric
(b) Process that is reversible and adiabatic: A reversible and adiabatic process is also called isentropic.
Answer for (b): C) Isentropic
(c) Process at constant pressure: A process at constant pressure is called an isobaric process.
Answer for (c): D) Isobaric
(d) Process at constant volume: A process at constant volume is isochoric.
Answer for (d): B) Isochoric
(e) Process at constant temperature: A process at constant temperature is isothermal.
Answer for (e): A) Isothermal

So the answers in sequence are:
(a) B
(b) C
(c) D

(d) B
(e) A
The correct option matching this sequence is E) B, C, D, B, A.

6. Consider a thermodynamic system characterized by the enthalpy H. Which of the following Maxwell's relations is valid for this system?
   A) $\left(\frac{\partial P}{\partial S}\right)_T = -\left(\frac{\partial T}{\partial V}\right)_P$
   B) $\left(\frac{\partial V}{\partial S}\right)_T = \left(\frac{\partial T}{\partial P}\right)_V$
   C) $\left(\frac{\partial V}{\partial T}\right)_S = \left(\frac{\partial S}{\partial P}\right)_V$
   D) $\left(\frac{\partial P}{\partial T}\right)_S = -\left(\frac{\partial S}{\partial V}\right)_P$
   E) $\boldsymbol{\left(\frac{\partial S}{\partial P}\right)_T = \left(\frac{\partial V}{\partial T}\right)_P}$ *(Erratum: Should be* $\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$ *, entire class has been awarded marks for this question)*

7. Consider an ideal gas defined by its enthalpy (H). For this system, the molar isothermal compressibility at constant temperature (T) is given as $\left(\frac{\partial S}{\partial P}\right)_T = 0.4\,\mathrm{J/(mol{\cdot}K{\cdot}Pa)}$ and the molar coefficient of thermal expansion at constant pressure (P) is $\left(\frac{\partial \underline{V}}{\partial T}\right)_P = 1.2 \times 10^{-6}\,\mathrm{m^3/mol{\cdot}K}$. Calculate the change in molar volume when the temperature of the system changes by 25 K during an isobaric (constant pressure) process. (Hint: Use the relations from the previous question)

   A) $2.5 \times 10^{-7}\ m^3/mol$
   B) $\mathbf{3.0 \times 10^{-7}\ m^3/mol}$ *(Erratum: Should be 3×10*$^{-5}$ *m*^{3}*/mol, entire class has been awarded marks for this question)*
   C) $6.0 \times 10^{-7}\ m^3/mol$
   D) $1.2 \times 10^{-6}\ m^3/mol$
   E) $2.4 \times 10^{-6}\ m^3/mol$

To solve this problem, we'll use the molar coefficient of thermal expansion $\left(\frac{\partial \underline{v}}{\partial T}\right)_P$, which is derived from the Maxwell relation for enthalpy (H).
The change in molar volume $d\underline{V}$ due to a temperature change dT at constant pressure can be calculated using:

$$d\underline{V} = \left(\frac{\partial \underline{V}}{\partial T}\right)_P dT$$

Given:

$$\left(\frac{\partial \underline{\underline{V}}}{\partial T}\right)_P = 1.2 \times 10^{-6}\ \mathrm{m^3/mol{\cdot}K}$$

$$\mathrm{d}T = 25\ \mathrm{K}$$

We find:

$$\mathrm{d}\underline{\underline{V}} = 1.2 \times 10^{-6}\ \mathrm{m^3/mol{\cdot}K} \times 25\ \mathrm{K} = 3.0 \times 10^{-5}\ \mathrm{m^3/mol}$$

The molar isothermal compressibility at constant temperature (T) is a red herring.

8. A cylinder contains a monoatomic ideal gas at an initial temperature of 300 K and an initial pressure of 2 atm. The gas undergoes adiabatic compression, and the final pressure is 8 atm. Considering $Cp=\frac{5}{2}R$, what is the final temperature of the gas?
   A) 1200 K
   B) 689 K
   C) 700 K
   D) 1010 K
   E) 900 K

Solution:
Given that T_1=300K, P_1 =2atm, P_2=8atm, and $Cp=\frac{5}{2}R$, we can rewrite the equation as:

$$\frac{T_2}{300\mathrm{K}} = \left(\frac{8\ atm}{2\ atm}\right)^{\frac{R}{\frac{5}{2}R}}$$

$$T2 \approx 522.33K$$

(Erratum: None of the solutions in the question are valid. Entire class has been awarded marks for this question.)

9. Which of the following is NOT a property of an ideal gas?

   A) The internal energy is not a function of the volume

   B) The enthalpy is not a function of the pressure

   C) PV = nRT

   D) $\boldsymbol{C_V - C_P = R}$

   E) No intermolecular forces

10. Which of the following statements best describes the concept of the "heat death" of the universe according to thermodynamic principles?

A) The universe will eventually reach a state of minimum entropy.

B) The universe will eventually contract into a singularity, releasing an enormous amount of heat.

C) The universe will eventually reach a state of maximum entropy where all matter is evenly distributed.

D) The universe will eventually reach thermal equilibrium but with decreasing entropy.

E) The universe will eventually collapse, leading to a decrease in overall entropy and increase in free energy.

The correct answer is C) The universe will eventually reach a state of maximum entropy where all matter is evenly distributed. This is consistent with the second law of thermodynamics, which states that the entropy of an isolated system tends to increase over time, leading to a state of maximum entropy. The "heat death" hypothesis suggests that the universe will reach a state where it is so spread out that it is essentially in thermal equilibrium, with no more room for work to be done.